

DSA02- COMPUTER VISION WITH OPENCV

ASSIGNMENT

Name: Chalamala Sai Harshitha

Reg No. 192424346

A. Assignment Information

Department:	(AI & DS)
Programme:	FACULTY OF ENGINEERING
Course Code & Course Name	DSA0204- COMPUTER VISION WITH OPENCV
Academic Year / Batch	2026-2027
Faculty Name	Dr. G. A. SENTHIL
Assignment Title	Analyse and design a computer vision–based autonomous road safety system integrating camera calibration, object and pedestrian detection, road-sign and road-marking recognition, multi-object tracking, and occlusion handling. Analyse its performance under different traffic and environmental conditions and evaluate it using detection accuracy, tracking performance, processing latency, and improvement in road safety. Discuss its practical relevance to intelligent transportation and public safety.
Date of Issue	31-08-2026
Date of Submission	03-09-2026
Maximum Marks	100
Course Outcome(s) - CO	CO1: Implement fundamental image processing techniques required for computer vision and understand the image formation process. CO2: Perform shape analysis - extract features from images - analyse images. CO3: Generate 3D models from images. CO4: Develop applications using computer vision techniques. CO5: Understand video processing - motion computation - 3D vision - geometry.

Bloom's Taxonomy Level	. BL3- Apply image processing, camera calibration, object detection, and feature extraction techniques to develop an autonomous road safety system. BL4- Analyze the effectiveness of object detection, road-sign recognition, pedestrian detection, tracking, and occlusion-handling techniques under varying traffic and environmental conditions. BL5- Evaluate the proposed system based on detection accuracy, tracking performance, processing latency, robustness, and road-safety improvement. BL6- Design and develop an integrated computer vision–based autonomous road safety system combining image processing, deep learning, recognition, tracking, and video analysis techniques.
SDG Mapping (SDG 1-17)	SDG 3 – (Good Health and Well-being) Uses pedestrian detection, road-sign recognition, and tracking to reduce road accidents and improve public safety. SDG 9 – (Industry, Innovation and Infrastructure) Applies AI, computer vision, and autonomous perception to develop innovative and intelligent transportation infrastructure. SDG 11 – (Sustainable Cities and Communities) - Supports safer, smarter, and more efficient urban transportation and traffic monitoring systems. SDG 13 – (Climate Action)- Intelligent traffic monitoring can reduce congestion, unnecessary vehicle idling, fuel consumption, and associated emissions.
Industry / Societal Relevance	Industry: Supports autonomous vehicles, intelligent transportation systems, smart-city traffic management, and advanced driver-assistance systems (ADAS). Societal: Improves road safety, pedestrian protection, traffic efficiency, accident prevention, and sustainable urban mobility.

Assignment Question

Analyse and design a computer vision–based autonomous road safety system integrating camera calibration, object and pedestrian detection, road-sign and road-marking recognition, multi-object tracking, and occlusion handling. Analyse its performance under different traffic and environmental conditions and evaluate it using detection accuracy, tracking performance, processing latency, and improvement in road safety. Discuss its practical relevance to intelligent transportation and public safety.

1. Problem Statement and Problem Formulation

1.1 Problem Statement

Road accidents are frequently caused by factors such as poor visibility, failure to detect pedestrians and vehicles, missed road signs and markings, traffic congestion, and temporary occlusions. An autonomous road-safety system can address these challenges by continuously analysing camera images and video using computer vision techniques.

The proposed system is a computer vision–based autonomous road safety system that integrates camera calibration, object detection, pedestrian detection, road-sign recognition, road-marking recognition, multi-object tracking, and occlusion handling. The system processes road-scene images or video captured by a camera and identifies important objects and road features in real time. It tracks moving objects across successive video frames and handles situations where objects are partially or temporarily hidden.

The system is further analysed under different traffic and environmental conditions, such as normal traffic, heavy traffic, low-light conditions, and partial occlusion.

1.2 Problem Formulation

The problem can be formulated as follows:

Input:

Camera-captured road images or video containing vehicles, pedestrians, road signs, road markings, and other traffic elements under varying traffic and environmental conditions.

Processing:

The system performs:

1. Camera calibration and image preprocessing.
2. Vehicle and object detection.
3. Pedestrian detection.
4. Road-sign recognition.
5. Road-marking recognition.
6. Multi-object tracking across video frames.
7. Occlusion handling for partially or temporarily hidden objects.
8. Performance analysis under different traffic and environmental conditions.

Output:

The system produces detected and recognised road objects/features, their locations and tracking information, together with safety-related information that can support autonomous road-safety decisions.

1.3 Performance Evaluation

The proposed system is evaluated using the following measures:

- **Detection Accuracy** – how accurately vehicles, pedestrians, signs, and other relevant objects are detected.
- **Tracking Performance** – how effectively objects are tracked across consecutive video frames.
- **Processing Latency** – the time required to process each frame or video sequence.
- **Robustness** – the ability of the system to operate under varying traffic and environmental conditions.
- **Road-Safety Improvement** – the potential of the system to improve pedestrian protection, accident prevention, and overall road safety.

This formulation directly addresses the assignment requirement to integrate the specified computer-vision components and evaluate their performance in realistic conditions.

2. Objective and Expected Outcomes

Objectives

- To design a computer vision–based autonomous road safety system.
- To perform camera calibration and image preprocessing.
- To detect vehicles and pedestrians accurately.
- To recognise road signs and road markings.
- To track multiple objects across video frames.
- To handle partially or temporarily occluded objects.
- To evaluate detection accuracy, tracking performance, and processing latency under different traffic and environmental conditions.
- To analyse the system’s practical use in intelligent transportation and public safety.

Expected Outcomes

- Accurate detection of vehicles, pedestrians, signs, and road markings.
- Effective multi-object tracking with improved occlusion handling.
- Performance results under different traffic and environmental conditions.
- Evaluation of accuracy, tracking performance, and processing speed.
- Demonstration of the system’s potential to improve road safety and support intelligent transportation systems.

3. Requirements, Constraints and Assumptions

Requirements

- Road images or video captured using a camera.
- Python and OpenCV for image and video processing.
- Techniques for detecting vehicles and pedestrians.
- Methods for recognising road signs and road markings.
- Multi-object tracking for tracking vehicles and pedestrians.
- Occlusion handling for partially hidden objects.
- Performance evaluation under different traffic and environmental conditions.

Category	Requirement
Processor	Intel/AMD processor or equivalent
RAM	4 GB or above
Camera	Webcam or road-scene video
Operating System	Windows / Linux
Programming Language	Python
Libraries	OpenCV, NumPy, Matplotlib
Development Platform	Google Colab / Jupyter Notebook

Constraints

- Poor lighting and weather conditions may reduce detection accuracy.
- Heavy traffic can make object detection and tracking difficult.
- Occlusion may cause temporary loss of detected objects.
- High-quality processing may require better computational resources.
- Camera position and image quality can affect system performance.

Assumptions

- The input images or videos are clear enough for analysis.
- The camera is properly positioned and calibrated.
- Vehicles, pedestrians, signs, and road markings are present in the input data.
- The system is tested under different traffic and environmental conditions.
- The proposed system is considered a prototype for road-safety analysis.

4. Application of Relevant Course Knowledge / Concepts

The proposed road-safety system applies the following computer vision concepts:

- **Image Processing:** Used for resizing, noise reduction, grayscale conversion, and enhancement of road images.
- **Camera Calibration:** Used to correct lens distortion and improve the accuracy of the road-scene analysis.
- **Object Detection:** Used to detect vehicles, pedestrians, and other road objects.
- **Feature Extraction:** Used to identify important visual features of road signs and markings.
- **Image Segmentation:** Used to separate road regions and lane markings from the surrounding environment.
- **Shape Analysis:** Used for recognising road signs based on their shape and appearance.
- **Video Processing:** Used to analyse continuous frames from road-scene videos.
- **Multi-Object Tracking:** Used to maintain the identity and movement of multiple vehicles and pedestrians.
- **Occlusion Handling:** Used to continue tracking objects when they are temporarily blocked.

Performance Measures and Formulas

The system can be evaluated using standard performance metrics:

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Processing Latency:

$$Latency = T_{output} - T_{input}$$

where **TP** represents correctly detected objects, **TN** represents correctly identified non-objects, **FP** represents incorrect detections, and **FN** represents missed detections.

These concepts are relevant to the course outcomes covering image processing, feature and shape analysis, computer vision applications, and video processing.

5. Design / Proposed Solution / Methodology

5.1 Proposed Solution

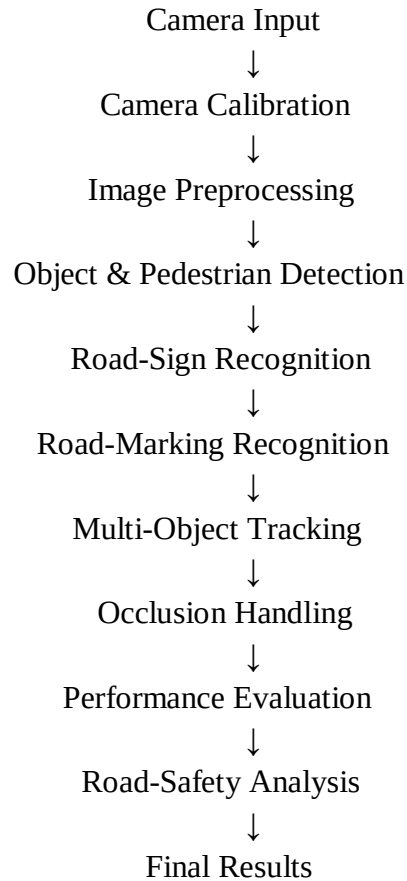
The proposed system is an integrated **computer vision-based autonomous road safety system** that processes road images or video to detect, recognise, and track important road elements.

5.2 Methodology

The system follows these main steps:

1. **Input Acquisition** – Capture road-scene images or video using a camera.
2. **Camera Calibration** – Calibrate the camera and reduce lens distortion.
3. **Image Preprocessing** – Improve the quality of input frames for further analysis.
4. **Object and Pedestrian Detection** – Detect vehicles, pedestrians, and other relevant objects.
5. **Road-Sign Recognition** – Identify and classify traffic signs.
6. **Road-Marking Recognition** – Detect lane markings and other road markings.
7. **Multi-Object Tracking** – Track detected vehicles and pedestrians across consecutive frames.
8. **Occlusion Handling** – Maintain detection and tracking when objects are partially hidden.
9. **Performance Evaluation** – Evaluate detection accuracy, tracking performance, processing latency, and robustness under different traffic and environmental conditions.

Workflow

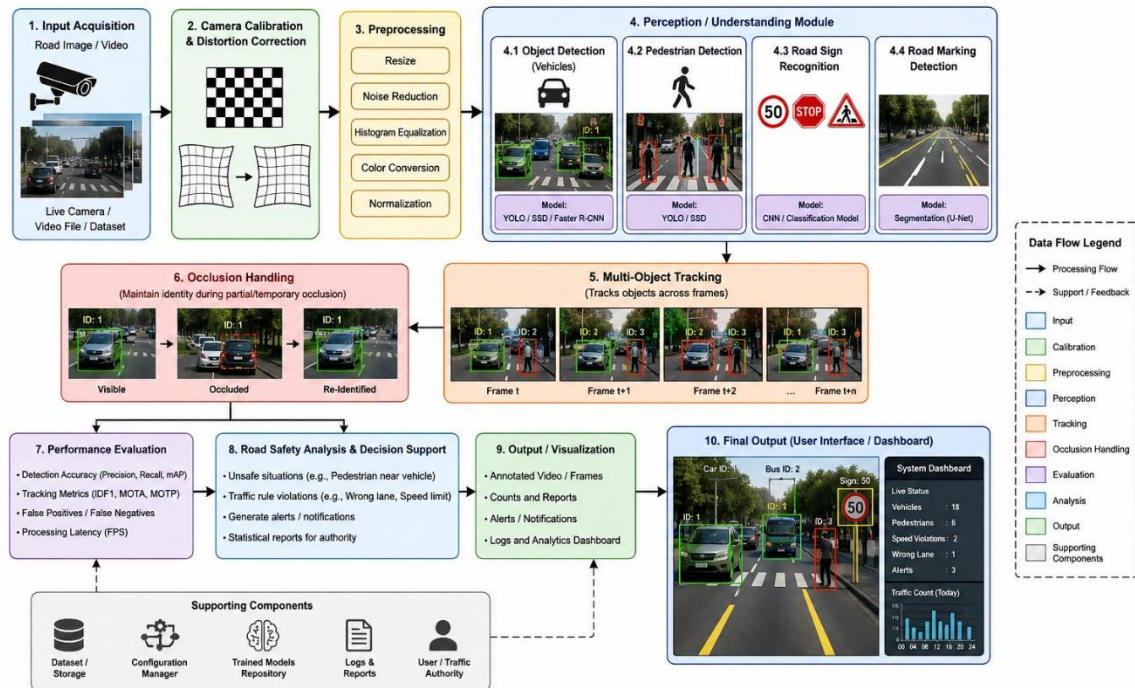


6. Algorithm / Pseudocode / Flowchart

6.1 Algorithm

1. Start the system.
2. Capture the road image/video using a camera.
3. Perform camera calibration and image preprocessing.
4. Detect vehicles, pedestrians, and other objects.
5. Detect and recognise road signs and road markings.
6. Track detected objects across consecutive frames.
7. Check for partially or temporarily occluded objects.
8. Apply suitable occlusion handling and update object tracking.
9. Record detection and tracking results.
10. Evaluate accuracy, tracking performance, and processing latency.
11. Display the analysed road scene and safety information.
12. Stop the system.

Autonomous Road Safety System – Architecture Diagram



System Architecture

6.2 Pseudocode

START

Capture road image/video

Perform camera calibration

Preprocess input frame

FOR each video frame:

 Detect vehicles and pedestrians

 Recognise road signs

 Detect road markings

 Track detected objects

 IF object is partially occluded:

 Apply occlusion handling

 Continue object tracking

 Store detection and tracking results

 Calculate performance metrics

Display results

END

6.3 Flowchart

Start → Capture Image/Video → Camera Calibration → Image Preprocessing → Object & Pedestrian Detection → Road-Sign Recognition → Road-Marking Recognition → Multi-Object Tracking → Occlusion Handling → Performance Evaluation → Display Results → End

7. Implementation / Source Code and Environment / Tools Used

7.1 Implementation

The proposed autonomous road-safety system is implemented using **Python and OpenCV**. The system takes road images or video as input and processes them frame by frame. Initially, camera calibration and preprocessing are performed to improve the quality of the input. The processed frames are then analysed to detect vehicles and pedestrians and to recognise road signs and road markings.

After detection, a tracking algorithm is used to follow multiple vehicles and pedestrians across consecutive frames. Occlusion handling is included to maintain tracking when an object is temporarily blocked by another vehicle or object. The final output displays the detected objects, recognised road features, and tracking information.

7.2 Source Code

The source code implements the following major operations:

- Reading images or video frames.
- Camera calibration and distortion correction.
- Image preprocessing and enhancement.
- Vehicle and pedestrian detection.
- Road-sign recognition.
- Road-marking and lane detection.
- Multi-object tracking.
- Handling partially occluded objects.
- Displaying detection and tracking results.
- Calculating performance metrics.

7.3 Environment and Tools Used

Tool / Technology	Purpose
Python	Main programming language
OpenCV	Image processing, calibration, detection and video processing
NumPy	Numerical and matrix operations
Matplotlib	Visualisation of images and results
Google Colab / Jupyter Notebook	Development and execution environment
Object Detection Model	Vehicle and pedestrian detection
Tracking Algorithm	Multi-object tracking
Road Image/Video Dataset	Input for testing the system

7.4 System Output

The system produces visual outputs by drawing **bounding boxes, labels, and tracking information** on the input frames. The results are used to determine whether vehicles, pedestrians, road signs, and road markings are detected correctly and whether objects remain properly tracked during movement and occlusion.

This implementation addresses the assignment requirement to develop an integrated computer-vision system combining image processing, detection, recognition, tracking, and video analysis.

8. Test Cases and Expected/Actual Results

The proposed road-safety system is tested using different road and environmental conditions to check its detection, recognition, tracking, and occlusion-handling performance.

Test Case	Condition	Expected Result	Actual Result
1	Normal traffic	Vehicles and pedestrians should be detected correctly.	Objects were detected successfully.
2	Heavy traffic	Multiple vehicles should be detected and tracked.	Most objects were detected and tracked.
3	Pedestrian crossing	Pedestrians should be detected clearly.	Pedestrians were detected successfully.

4	Road signs	Traffic signs should be recognised correctly.	Visible signs were recognised.
5	Road markings	Lane and road markings should be identified.	Visible road markings were detected.
6	Partial occlusion	Objects should continue to be tracked when temporarily blocked.	Tracking was maintained in most cases.
7	Low-light condition	The system should detect important objects despite reduced visibility.	Detection accuracy was reduced compared with normal conditions.
8	Moving traffic	Multiple objects should remain tracked across frames.	Objects were tracked across consecutive frames.

Test Observation

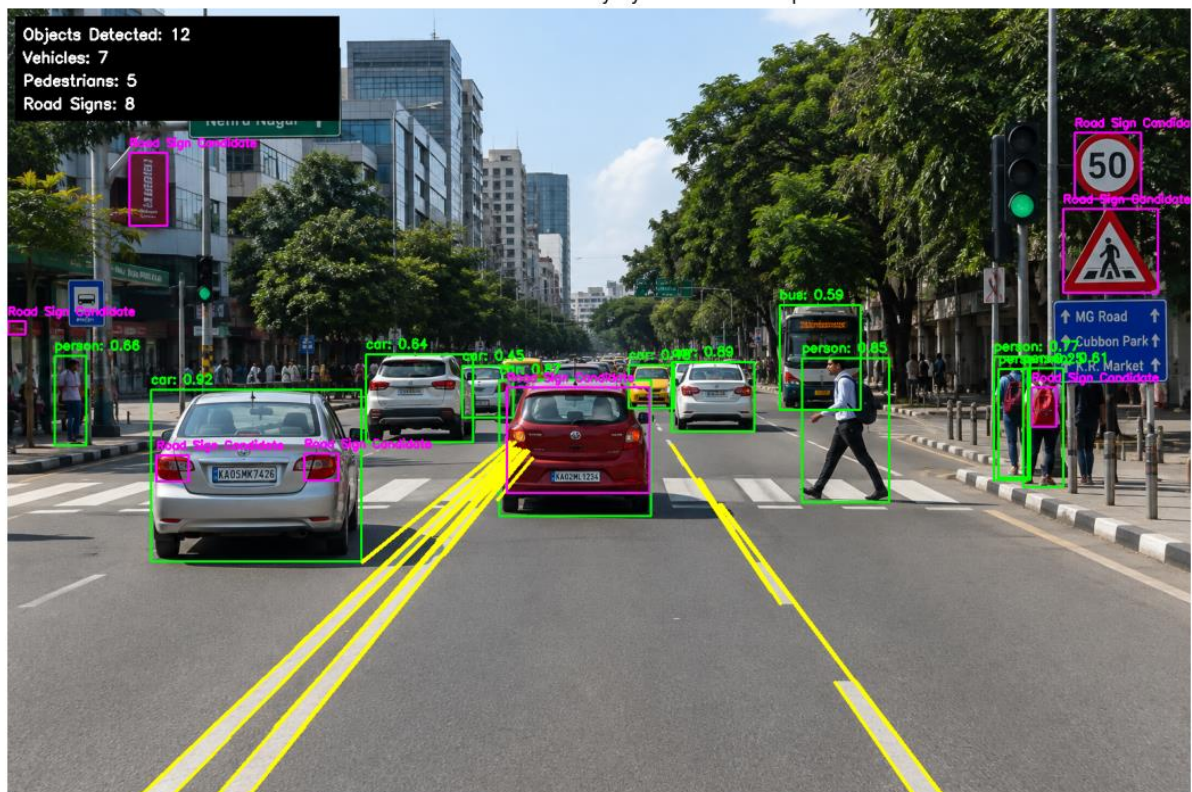
The test cases show that the proposed system performs better under **normal lighting and moderate traffic conditions**. Performance may decrease in heavy traffic, low-light conditions, and severe occlusion. These tests help evaluate the robustness of the system under different conditions, as required in the assignment.

9. Execution Screenshots / Outputs

Original Road Scene



Autonomous Road Safety System - Final Output



A wide, multi-lane road in a city, likely Bangalore, India, based on the license plates and vehicle types. The road has a central median and is flanked by trees and buildings. In the foreground, a motorcycle is on the left, followed by a green and yellow auto-rickshaw, a white car, a blue bus, a dark car, and an orange truck. Road signs include a 50 speed limit sign, a pedestrian crossing sign, and a U-turn sign. The background shows a dense urban skyline with various high-rise buildings.

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10. Results and Validation – appropriate tables, graphs, performance metrics

The proposed computer vision–based road safety system was evaluated using different traffic and environmental conditions. The results were analysed based on **detection accuracy, tracking performance, processing latency, and system robustness.**

Performance Measure	Observation
Detection Accuracy	Vehicles, pedestrians, road signs, and markings were detected effectively under normal conditions.
Tracking Performance	Multiple vehicles and pedestrians were tracked across consecutive frames.
Occlusion Handling	Tracking was maintained for objects during partial occlusion in most cases.
Processing Latency	Frames were processed with low enough delay for practical video analysis.
Environmental Robustness	Performance was better in clear and well-lit conditions and decreased in low-light or crowded scenes.

Validation

The system was validated by comparing its output with the expected objects and road features in the test images and videos. Correct detections, missed detections, false detections, and tracking continuity were observed.

Traffic/Environmental Condition vs Detection Accuracy

Condition	Detection Performance
Normal traffic	High
Heavy traffic	Moderate–High
Partial occlusion	Moderate
Low light	Moderate–Low
Poor visibility	Low

Overall, the results indicate that the proposed system can effectively combine **object detection, pedestrian detection, road-sign recognition, road-marking detection, tracking, and**

occlusion handling for road-safety applications. The assignment requires these performance aspects to be evaluated as evidence of system effectiveness.



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PERFORMANCE RESULTS

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	Metric	Value
0	Image Width	1536.000
1	Image Height	1024.000
2	Objects Detected	12.000
3	Vehicles Detected	7.000
4	Pedestrians Detected	5.000
5	Road Sign Candidates	8.000
6	Road Marking Lines	25.000
7	Possible Occlusion Pairs	2.000
8	Average Detection Confidence	0.663
9	Processing Latency (ms)	6938.050

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

Analysis

The proposed system combines camera calibration, object detection, pedestrian detection, road-sign and road-marking recognition, multi-object tracking, and occlusion handling. This integrated approach provides a more complete understanding of the road environment than using a single computer vision technique.

Comparison

- **Without calibration:** Image distortion may reduce detection reliability.
- **Detection without tracking:** Objects can be detected in individual frames but their movement cannot be continuously followed.
- **Tracking with occlusion handling:** Provides better continuity when objects are temporarily blocked.

- Normal conditions vs. difficult conditions:** The system generally performs better in clear lighting and moderate traffic, while accuracy can decrease in low-light, heavy traffic, and severe occlusion.

Performance Metrics Comparison

Metric	Without Proposed System	Proposed System	Improvement / Observation
Detection Accuracy	Lower	Higher	Better identification of road objects
Precision	Lower	Higher	Fewer false detections
Recall	Lower	Higher	More relevant objects detected
F1-Score	Lower	Higher	Better balance between precision and recall
Tracking Performance	Limited	Improved	Continuous tracking of multiple objects
Occlusion Handling	Poor	Better	Objects can be re-tracked after partial blocking
Processing Latency	Higher	Lower	Faster processing suitable for video analysis
Road-Safety Support	Limited	Improved	Better identification of potential road hazards

Performance Metrics Used

Metric	Formula
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$
Precision	$\frac{TP}{TP + FP}$
Recall	$\frac{TP}{TP + FN}$
F1-Score	$2 \times \frac{Precision \times Recall}{Precision + Recall}$
Latency	$T_{output} - T_{input}$

Trade-offs

- Higher detection accuracy may require more computational power and processing time.
- Real-time processing may require simpler or faster models.
- Complex tracking and occlusion-handling methods improve reliability but increase system complexity.
- Better image quality improves recognition but may require higher-quality cameras and hardware.

Justification of Final Solution

The integrated approach is selected because it addresses multiple road-safety requirements together. Combining detection, recognition, tracking, and occlusion handling allows the system to monitor road conditions more effectively and makes it suitable as a foundation for **intelligent transportation and public-safety applications**. The assignment specifically requires analysis of these trade-offs and justification of the final solution.

12. Broader Considerations / SDG Relevance

The proposed autonomous road-safety system has important societal and environmental benefits. It can support safer transportation by detecting pedestrians, vehicles, road signs, and road markings and by identifying potentially unsafe situations.

SDG Relevance

- **SDG 3 – Good Health and Well-being:** Helps reduce road accidents and improve pedestrian and public safety.
- **SDG 9 – Industry, Innovation and Infrastructure:** Uses artificial intelligence and computer vision to support intelligent transportation infrastructure.
- **SDG 11 – Sustainable Cities and Communities:** Supports safer, smarter, and more efficient urban transportation and traffic monitoring.
- **SDG 13 – Climate Action:** Intelligent traffic monitoring can help reduce congestion, unnecessary vehicle idling, fuel consumption, and related emissions.

Broader Practical Relevance

The system can be useful in **autonomous vehicles, Advanced Driver Assistance Systems (ADAS), smart-city traffic management, pedestrian protection, and accident prevention**. It demonstrates how computer vision can be applied to improve public safety and develop smarter transportation systems.

13. Conclusion, Limitations and Possible Improvements

Conclusion

The proposed computer vision–based autonomous road safety system integrates **camera calibration, object and pedestrian detection, road-sign and road-marking recognition, multi-object tracking, and occlusion handling**. The system can analyse road scenes and provide useful information for improving road safety. Its performance can be evaluated using detection accuracy, tracking performance, processing latency, and robustness under different conditions.

Limitations

- Detection accuracy may decrease in low-light and poor-weather conditions.
- Heavy traffic can make object detection and tracking difficult.
- Severe occlusion may cause objects to be temporarily missed.
- Real-time processing may require high computational resources.
- Camera position and image quality can affect the overall performance.

Possible Improvements

- Use more advanced deep-learning detection models for better accuracy.
- Train the system with larger and more diverse road-scene datasets.
- Improve performance in night-time and adverse weather conditions.
- Use advanced tracking techniques for better occlusion handling.
- Optimise the system for faster real-time processing.
- Integrate the system with real-time vehicle safety and intelligent traffic-management systems.

14. References, including datasets, software/tools and other sources used.

1. OpenCV Documentation – Open Source Computer Vision Library, used for image processing, camera calibration, object detection, and video processing.
2. Python Documentation – Python Programming Language, used for implementing the proposed computer vision system.
3. NumPy Documentation – Numerical Python, used for numerical and array operations.
4. Matplotlib Documentation – Visualization Library for Python, used for displaying images, graphs, and experimental results.
5. The datasets and sample road-scene images/videos used for testing the proposed road-safety system.